

Assignment - 2

Course Name: Algebraic Geometry

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Throughout, k denotes an algebraically closed field unless otherwise specified.

- (1) Let $X \subset \mathbb{A}_k^n$ be a quasi-affine variety and $\bar{X}$ be its closure in $\mathbb{A}_k^n$. Prove that $\dim X = \dim \bar{X}$.
- (2) If we identify $\mathbb{A}_k^2$ with $\mathbb{A}_k^1 \times \mathbb{A}_k^1$ in the natural way as sets, show that the Zariski topology on $\mathbb{A}_k^2$ is not the product topology of the Zariski topologies on the two copies of $\mathbb{A}_k^1$.
- (3) Let C be an affine conic (i.e., an affine irreducible plane curve of degree 2). Show that C is isomorphic to the plane curve $\{y - x^2 = 0\}$ or $\{xy - 1 = 0\}$ in $\mathbb{A}_k^2$ with respect to the affine coordinates x and y of $\mathbb{A}_k^2$.

- (4) Consider the *twisted cubic curve* $C \subset \mathbb{A}_k^3$ parametrically represented (using a single parameter t) as

$$x = t, \quad y = t^2, \quad z = t^3,$$

where x, y, z are the affine coordinates of $\mathbb{A}_k^3$. Show that C is isomorphic to $\mathbb{A}_k^1$.

- (5) Let X be the algebraic set in $\mathbb{A}_k^3$ defined by the two polynomials

$$x^2 - yz \quad \text{and} \quad xz - x.$$

Show that X is a union of three irreducible components. Describe them and find their prime ideals.

- (6) (a) If Y is any subset of a topological space X , show that $\dim Y \leq \dim X$.
 (b) If X is a topological space which has an open covering $\{U_i\}_{i \in \Lambda}$, then show that

$$\dim X = \sup_{i \in \Lambda} \dim U_i.$$

- (c) Give an example of a topological space X and a dense open subset U with $\dim U < \dim X$.
- (d) If Y is a closed subset of an irreducible finite-dimensional topological space X , and if $\dim Y = \dim X$, then show that $Y = X$.
- (e) Can there be a noetherian topological space of infinite dimension?

- (7) Prove that the projective n -space $\mathbb{P}_k^n$ is a noetherian topological space which is also irreducible.

- (8) Let $X \subset \mathbb{P}_k^n$ be a quasi-projective variety and $\bar{X}$ be its closure in $\mathbb{P}_k^n$. Prove that $\dim X = \dim \bar{X}$.

- (9) Prove that $\mathbb{A}_k^1$ is not isomorphic to any proper Zariski open subset of itself.

- (10) (a) Show that $\mathbb{A}_k^2$ is not homeomorphic to $\mathbb{P}_k^2$ with respect to their Zariski topologies.
 (b) Show that if an affine variety is isomorphic to a projective variety, then it consists of only one point.
- (11) Let $f \in k[X_0, \dots, X_n]$ be a homogeneous polynomial. If g is an irreducible factor of f , prove that g is again homogeneous.

- (12) Let P be a point on a variety (affine, or quasi-affine, or projective, or quasi-projective) X . Show that $\dim \mathcal{O}_{X,P} = \dim X$.

- (13) Let $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n)$ be two distinct points of $\mathbb{A}_k^n$.
 (a) Define the line $\overline{AB}$ passing through A and B .

- (b) Show that if L is the line $\overline{AB}$ joining A and B , and $T : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ is an affine change of coordinates, then $T(L)$ is the line through the points $T(A)$ and $T(B)$.
- (14) Let $A = (a_0 : a_1 : a_2)$ and $B = (b_0 : b_1 : b_2)$ be distinct points in $\mathbb{P}_k^2$. Give an equation for the line $\overline{AB}$ passing through A and B .
- (15) Let P_1, P_2, P_3 (respectively Q_1, Q_2, Q_3) be three non-collinear points in $\mathbb{P}_k^2$. Show that there is a projective change of co-ordinates $T : \mathbb{P}_k^2 \rightarrow \mathbb{P}_k^2$ such that $T(P_i) = Q_i$, for $i = 1, 2, 3$.
- (16) Let $H_1, \dots, H_m$ be hyperplanes in $\mathbb{P}_k^n$ with $m \leq n$. Show that $H_1 \cap H_2 \cap \dots \cap H_m \neq \emptyset$.